CMT Lab

Individual Contribution Report

1. Module Overview

The broker-front-end is an Angular-based user interface for the broker side of the FIX Trading Simulator. It enables users to submit, monitor, and cancel orders in real time.

The frontend communicates with the backend using REST APIs for order actions and WebSocket connections for real-time updates.

2. Assigned Responsibilities

- Implement Cancel Order workflow from UI to backend
- Handle server responses and error scenarios
- Ensure UI consistency with asynchronous updates
- Improve resilience to connection issues
- Manage and display order states in the UI

3. Analysis of Existing Codebase

3.1 Existing Features

The system already provided a base implementation for order handling, which was extended and improved.

- Basic cancelOrder() using REST API
- WebSocket connection with auto-reconnect
- Order status display (NEW, FILLED, CANCELED, REJECTED)
- Polling mechanism for refreshing orders
- Basic error display in UI

3.2 Identified Gaps

- Cancel flow not aligned with real-time updates
- No intermediate UI feedback during cancel

- No UI indication of connection state
- Limited error visibility
- No structured tracking of order events

4. Implementation Details

4.1 Cancel Order Workflow

- Added Cancel button in UI
- Triggered backend cancel via REST API
- Handled responses using WebSocket and polling
- Ensured UI reflects updated order state

4.2 WebSocket Integration

- Handled real-time order updates
- Implemented reconnection logic
- Maintained UI synchronization with backend

4.3 Handling Asynchronous Updates

- Combined WebSocket updates with polling
- Ensured consistency during delayed responses

4.4 Error Handling

- Displayed user-friendly error messages
- Handled failed cancel scenarios gracefully

4.5 UI State Management

- Managed order states (NEW, FILLED, CANCELED, REJECTED)
- Applied visual styles for clarity

5. Files Contributed

- order.service.ts — WebSocket + API handling
- app.component.ts — Cancel workflow + logic
- app.component.html — UI rendering

- app.component.scss — Styling

File	Contribution
order.service.ts	Implemented WebSocket handling, real-time updates, and reconnection logic; integrated backend API calls.
app.component.ts	Handled cancel order workflow, API interaction, error handling, and UI state updates.
app.component.html	Developed UI elements including cancel button, order display, and error messages.
app.component.scss	Styled order states and improved visual clarity of UI components.

6. Key Design Decisions

- REST for actions, WebSocket for updates
- Polling as fallback for reliability
- Backend as source of truth

7. Testing & Validation

- Tested cancel order functionality
- Verified error handling
- Tested WebSocket reconnection
- Validated UI consistency under delays

Conclusion

The implementation delivers a reliable UI for order management by combining REST and WebSocket approaches while ensuring consistency and resilience.

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